Integer quotients of ternary Cantor numbers: a power lower bound and an infinite exceptional family

Timofei Iuzhakov

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Abstract

Let $\mathcal{D}$ be the nonnegative integers whose ternary digits belong to $\{0,1\}$. We prove two results about positive integer quotients of members of $\mathcal{D}$. First, the number $M_{\mathrm{C}}(X)$ of integers $1 \le M \le X$ with $N, MN \in \mathcal{D}$ for some positive integer N satisfies

$$M_{\mathrm{C}}(X) \gg X^{\beta_*}, \qquad \beta_* = \frac{(1+s)^2}{7-5s} = 0.691726633148796\ldots, \quad s = \log_3 2.$$

Version 2 improves the original exponent $0.6660585297\ldots$, without an epsilon loss, by an endpoint mean-discrepancy estimate and an exact fixed-residue-class energy certificate. The same bound holds for the number of positive-dimensional intersections of the 3-adic Cantor set with its inverse integer dilates. Second, for every $k \ge 0$, the integer $621 \cdot 3^{4k} - 20$ is not such a quotient, although it avoids the standard lowest-digit and leading-digit obstructions. This proves that the sporadic exception set of Bai, Meleshko, Riasat and Shallit is infinite. The first proof combines Fourier sampling over Cantor-restricted denominators with a finite-word Fibonacci estimate and a sum over 3-adic distances. The second constructs an explicit predecessor-closed carry certificate of $38k + 50$ states for $k \ge 1$, with a separate certificate for $k = 0$. The original proofs and the new improvement are given in full.

1 The quotient problem and the two main results

Version history. Version 1 (6 November 2026) established Theorems 1.1 and 1.2. Version 2 (8 November 2026) retains their full proofs and adds Theorem 9.1 in Section 9, raising the exponent from $0.6660585297\ldots$ to $0.691726633148796\ldots$ without an epsilon loss. The article title, sole author, and citation key `iuzhakov2026cantorquotients` are unchanged. The sharp growth of M_{C} and Erdős Problem 406 remain open here.

Put

$$\mathcal{D} = \left\{ \sum_{j \ge 0} \epsilon_j 3^j : \epsilon_j \in \{0,1\},\ \epsilon_j = 0 \text{ for all large } j \right\}, \qquad \mathcal{D}^+ = \mathcal{D} \setminus \{0\},$$

and let

$$\Sigma = \left\{ \sum_{j \ge 0} \epsilon_j 3^j : \epsilon_j \in \{0,1\} \right\} \subset \mathbb{Z}_3.$$

Here $\mathbb{N}$ denotes the positive integers. Define the set of positive integer quotients and its counting function by

$$\mathcal{V} = \mathbb{N} \cap (\mathcal{D}^+/\mathcal{D}^+), \qquad M_{\mathrm{C}}(X) = \#(\mathcal{V} \cap [1, X]) \quad (X \ge 1).$$

Equivalently, $M \in \mathcal{V}$ if some positive integer N has $N, MN \in \mathcal{D}$. Positivity of the witness is essential: allowing $N = 0$ would make this condition vacuous. For a positive integer M, also put

$$C(1, M) = \{x \in \mathbb{Z}_3 : x, Mx \in \Sigma\}, \qquad M_{\mathrm{H}}(X) = \#\{1 \le M \le X : \dim_H C(1, M) > 0\}.$$

1.1 A power lower bound

Lagarias asks for upper and lower bounds for these two counting functions in Problems 1.10 and 1.11 of [1], numbered Problems 1 and 2 in the arXiv version. Taking $N = 1$ gives the immediate lower bound of order $X^{\log_3 2}$. A positive integer witness also implies positive Hausdorff dimension by the nonoverlapping-block construction in [1]; we include this construction below. The inclusion of the integer-witness multipliers in the positive-dimensional multipliers is strict in general.

Theorem 1.1. *Let*

$$s = \log_3 2, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad g = \log_3 \varphi, \qquad \alpha = \frac{s-g}{1+g}, \qquad \beta = s + (2s-1)\alpha.$$

There are absolute constants $c > 0$ and X_0 such that

$$M_{\mathrm{C}}(X) \ge cX^{\beta}, \qquad M_{\mathrm{H}}(X) \ge cX^{\beta} \quad (X \ge X_0).$$

In particular,

$$\alpha = 0.1341512347\ldots, \qquad \beta = 0.6660585297\ldots > \log_3 2.$$

The result improves the exponent in the immediate lower bound. It does not determine the sharp asymptotic size or positive density of either counting function. The distribution estimates used in its proof are proved directly, with their dependence on growing denominators explicit.

1.2 An infinite exceptional family

Following Bai, Meleshko, Riasat and Shallit [4, §1.4], define

$$\begin{aligned}
\mathcal{O}_3 &= \{N \in \mathbb{N} : N \equiv 2 \cdot 3^{i-1} \pmod{3^i} \text{ for some } i \ge 1\},\\
\mathcal{O}_\infty &= \mathbb{N} \cap \bigcup_{i \ge 0} [\tfrac{3}{2}\, 3^i,\ 2 \cdot 3^i],\\
\mathcal{F} &= \mathbb{N} \setminus (\mathcal{V} \cup \mathcal{O}_3 \cup \mathcal{O}_\infty).
\end{aligned}$$

The first two sets encode the lowest nonzero digit and leading-digit obstructions to being a quotient. They are denoted $\mathcal{D}$ and $\mathcal{E}$ in [4]; our different symbols avoid conflict with $\mathcal{D}$, the digit-integer set. The original Cantor integers in that paper have digits $0, 2$. Their positive members form $2\mathcal{D}^+$, so their positive quotient set is exactly $\mathcal{D}^+/\mathcal{D}^+$. The sporadic set $\mathcal{F}$ therefore agrees with theirs.

Bai et al. ask whether $\mathcal{F}$ is infinite and propose the family in the following theorem.

Theorem 1.2. *For every integer $k \ge 0$,*

$$621 \cdot 3^{4k} - 20 \in \mathcal{F}.$$

In particular, there are no $a, b \in \mathcal{D}^+$ with $b = (621 \cdot 3^{4k} - 20)a$, and $\mathcal{F}$ is infinite.

The case $k = 4$ is verified in Jandhyala's versioned preprint [8, §6], which leaves the all-k assertion open. Its main theorems concern the different quotient problem for zero-free balanced digits $\{-1, 1\}$. Our proof of Theorem 1.2 uses the identity $b + 20a = 23 \cdot 3^{4k+3}a$ to obtain a binary delay register with a carry in a fixed set of 22 integers. An explicit predecessor-closed set separates the initial state from the only nontrivial predecessor of an accepting state.

1.3 Relation between the two arguments

The quotient problem is also studied directly by Anders, Dawsey, Reznick and Sisneros-Thiry [5]. Multiplicative intersections and their automata are treated in [2, 3]. The analytic part of our counting proof is related to work on the distribution of integers with missing digits [6, 7], but no external uniformity estimate is used without proof.

The two main theorems are logically independent. The first proves abundance among all positive integers; the second gives specified integers which fail to be quotients despite the elementary necessary conditions. The exceptional family is not a matching upper bound for $M_{\mathrm{C}}(X)$, and its nonrepresentation does not by itself determine $\dim_H C(1, M)$ for those multipliers.

Neither theorem concerns the number of exponents n for which $2^n \in \mathcal{D}$. In particular, neither proves the pointwise assertion in Erdős problem 406, excludes 1 from the infinite-visit exceptional set, or determines a finite-visit dimension limit. The arguments below use neither a general causal-container theorem nor a finite-rank unit-equation theorem. We make no claim of exhaustive priority verification.

2 Fourier sampling over reduced fractions

For $l \ge 0$, let

$$\mathcal{D}_l = \left\{ \sum_{j=0}^{l-1} \epsilon_j 3^j : \epsilon_j \in \{0,1\} \right\}, \quad F_l(t) = 2^{-l} \sum_{z \in \mathcal{D}_l} \mathrm{e}(zt), \quad f_l(t) = |F_l(t)|,$$

where $\mathrm{e}(t) = \exp(2\pi i t)$ and $t \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$. The product identity gives

$$F_l(t) = \prod_{j=0}^{l-1} \frac{1 + \mathrm{e}(3^j t)}{2}, \qquad f_{l+l'}(t) = f_l(t) f_{l'}(3^l t), \qquad 0 \le f_l(t) \le 1. \tag{1}$$

Lemma 2.1 (Elementary sampling)**.** *Let $0 < \delta \le 1$ and let $T \subset \mathbb{T}$ be finite, with circular separation at least δ. If $P(t) = \sum_{j=0}^{K} c_j \mathrm{e}(jt)$, then*

$$\sum_{t \in T} |P(t)|^2 \le (\delta^{-1} + 4\pi K) \sum_{j=0}^{K} |c_j|^2.$$

Proof. Center pairwise disjoint half-open arcs I_t of length δ at the points of T. For $G = |P|^2$ and $u \in I_t$,

$$G(t) \le G(u) + \int_{I_t} |G'(v)|\, dv.$$

Integrating in u and summing in t bounds the left side by $\delta^{-1}\|P\|_2^2 + \int_{\mathbb{T}} |G'|$. Since $\int |G'| \le 2\|P\|_2 \|P'\|_2 \le 4\pi K \|P\|_2^2$, Parseval proves the assertion. □

Lemma 2.2. *For every $h \geq 0$,*

$$\int_{\mathbb{T}} |F_h|^2 = 2^{-h}, \qquad \int_{\mathbb{T}} |F_h|^4 = (3/8)^h.$$

The degrees of F_h and F_h^2 are $(3^h - 1)/2$ and $3^h - 1$.

Proof. The second moment follows from the distinct frequencies in $\mathcal{D}_h$. For the fourth moment, the coefficients of F_h^2 count sums of two elements of $\mathcal{D}_h$. At each ternary position the sums $0, 1, 2$ have multiplicities $1, 2, 1$. There is no carry, so the sum of the squares of the unnormalized coefficients is $(1^2 + 2^2 + 1^2)^h = 6^h$. The normalization is 16^h. The degree statements follow from $\max \mathcal{D}_h = (3^h - 1)/2$. □

For real $D \geq 2$, write

$$T_D = \{v/d \bmod 1 : 2 \leq d \leq D,\ (d, 3) = 1,\ 1 \leq v < d,\ (v, d) = 1\}.$$

Distinct points of T_D have circular separation at least D^{-2}.

Lemma 2.3 (Farey Fourier bound)**.** *Put $\eta = \log_3(27/16) < s$. There is an absolute constant C such that, if $D \geq 2$ and $L \geq 3\lceil 2 \log_3 D \rceil$, then*

$$\sum_{t \in T_D} f_L(t) \leq C D^{\eta}. \tag{2}$$

For each fixed D, the sum on the left tends to zero as $L \to \infty$.

Proof. Take $h = \lceil 2 \log_3 D \rceil$, so $D^2 \leq 3^h < 3D^2$. Lemmas 2.1 and 2.2 give

$$\sum_{t \in T_D} f_h(t)^2 \leq C(3/2)^h, \qquad \sum_{t \in T_D} f_h(t)^4 \leq C(9/8)^h.$$

Cauchy–Schwarz yields

$$\sum_{t \in T_D} f_h(t)^3 \leq C(27/16)^{h/2} \leq C' D^{\eta}. \tag{3}$$

Let $p = \lfloor L/h \rfloor \geq 3$. By (1), after discarding the remainder of length $L - ph$, Hölder's inequality gives

$$\begin{aligned} \sum_{t \in T_D} f_L(t) &\leq \prod_{j=0}^{p-1} \left(\sum_{t \in T_D} f_h(3^{jh} t)^p \right)^{1/p} \\ &= \sum_{t \in T_D} f_h(t)^p \leq \sum_{t \in T_D} f_h(t)^3. \end{aligned}$$

Multiplication by 3^{jh} permutes T_D, since its reduced denominators are coprime to 3. This proves (2). For a fixed nonzero $v/d \in T_D$, the orbit under multiplication by 3 is periodic and avoids zero. The product in (1) therefore tends to zero in modulus. Summing over the fixed finite set proves the last assertion. □

3 Distribution for most Cantor denominators

For $Y \geq 3$, set

$$S_Y = \{N \in \mathcal{D} : 1 \leq N \leq Y,\ N \equiv 1 \pmod 3\}.$$

Elementary digit counting gives

$$\tfrac{1}{4} Y^s \leq |S_Y| \leq 2Y^s. \tag{4}$$

For the lower bound use all words of length $\lfloor \log_3 Y \rfloor$ whose low digit is 1. For the upper bound use all words of length $K = \lfloor \log_3 Y \rfloor + 1$.

Lemma 3.1 (Divisor multiplicity)**.** *For integers* $1 \le d \le Y$*,*

$$T_Y(d) := \#\{N \in S_Y : d \mid N\} \le 4(Y/d)^s.$$

Proof. Put $j = \lfloor \log_3 d \rfloor$. Fixing the digits at positions $j, \dots, K-1$ covers $\mathcal{D}_K$ by 2^{K-j} intervals, each of length $(3^j - 1)/2 < d$. Each interval contains at most one multiple of d. The inequalities $3^K \le 3Y$ and $3^j > d/3$ give $2^{K-j} < 4(Y/d)^s$. □

For $3 \nmid N$, define

$$R_N(L) = \sum_{v=1}^{N-1} f_L(v/N), \qquad R_1(L) = 0.$$

Proposition 3.2 (Average discrepancy on restricted denominators)**.** *As* $Y, L \to \infty$ *subject to* $L \ge 3\lceil 2\log_3 Y \rceil$*,*

$$\sum_{N \in S_Y} R_N(L) = o(Y^s). \tag{5}$$

Consequently, for all sufficiently large such Y, L*, at least* $Y^s/8$ *elements* $N \in S_Y$ *satisfy* $R_N(L) \le 1/2$*.*

Proof. Reducing the fractions v/N gives the exact identity

$$\sum_{N \in S_Y} R_N(L) = \sum_{\substack{2 \le d \le Y \\ (d,3)=1}} T_Y(d) \sum_{\substack{1 \le v < d \\ (v,d)=1}} f_L(v/d).$$

Use dyadic bins $(2^{j-1}, 2^j]$ for $1 \le j \le J = \lfloor \log_2 Y \rfloor$, followed by the possibly empty bin $(2^J, Y]$. By Lemmas 3.1 and 2.3, the contribution of a complete bin, divided by Y^s, is bounded by

$$C 2^{-js} \sum_{t \in T_{2^j}} f_L(t) \le C' 2^{-j(s-\eta)}.$$

The latter bounds are summable, since $\eta < s$. For each fixed j, the actual contribution tends to zero, by the last assertion of Lemma 2.3. First make the tail of the summable bounds small, and then take L large enough for the finitely many remaining bins. The final partial bin is at most $CY^{\eta - s}$ after division by Y^s, and also tends to zero. This proves (5). At most $2\sum_N R_N(L) = o(Y^s)$ denominators have $R_N(L) > 1/2$; combine this with (4). □

Proposition 3.3 (First moment)**.** *Let* $L = \lfloor \log_3 X \rfloor$*, and suppose* $Y \to \infty$ *with* $L \ge 3\lceil 2\log_3 Y \rceil$ *and* $X^s Y^{s-1} \to \infty$*. There is a subset* $G \subset S_Y$ *such that*

$$\sum_{N \in G} |A_N(X)| \ge \tfrac{1}{64} X^s Y^{2s-1}, \qquad A_N(X) = \{1 \le M \le X : NM \in \mathcal{D}\},$$

for all sufficiently large X*.*

Proof. Choose $G = \{N \in S_Y : R_N(L) \le 1/2\}$, which has size at least $Y^s/8$. For each $N \in G$ put $L_N = \lfloor \log_3(NX) \rfloor \ge L$. By (1), $R_N(L_N) \le R_N(L) \le 1/2$. Orthogonality of additive characters modulo N gives

$$\#\{z \in \mathcal{D}_{L_N} : N \mid z\} \ge \frac{2^{L_N}}{N}(1 - R_N(L_N)) \ge \frac{2^{L_N - 1}}{N}.$$

Every such z is at most NX. Removing zero and dividing by N yields

$$|A_N(X)| \ge \tfrac{1}{4} X^s N^{s-1} - 1.$$

Since $s < 1$, summing and using (4) gives

$$\sum_{N \in G} |A_N(X)| \ge \tfrac{1}{32} X^s Y^{2s-1} - 2Y^s \ge \tfrac{1}{64} X^s Y^{2s-1}.$$

□

4 A finite Fibonacci bound for overlaps

For $r \in 1 + 3\mathbb{Z}_3$, $r \neq 1$, put $e = v_3(r-1) \geq 1$. Define

$$W_n(r) = \{x \in \mathcal{D}_n : rx \bmod 3^n \in \mathcal{D}_n\},$$

using representatives in $[0, 3^n)$ for reduction. Let $H_n(e)$ be the number of independent sets of the graph on $\{0, \ldots, n-1\}$ whose edges join positions at distance e.

Lemma 4.1 (Finite-word bound)**.** *For every $n \geq 0$,*

$$|W_n(r)| \leq H_n(e) \leq \varphi^{n+e}.$$

The bound applies to all locally admissible words, including those which have no infinite admissible continuation.

Proof. Write $r = 1 + 3^e u$, where $u \in \mathbb{Z}_3^\times$. Write the input digits as $x_j \in \{0, 1\}$ and the digits of ux as $y_j \in \{0, 1, 2\}$. In the addition $x + 3^e ux$, let $\gamma_q \in \{0, 1\}$ be the incoming carry at position q. For $q \geq e$, put $B_q = y_{q-e} + \gamma_q$. The allowed values and next carries are exactly

B_q	allowed x_q	γ_{q+1}
0	0, 1	0
1	0	0
2	1	1
3	0, 1	1

Thus $\gamma_{q+1} = \mathbf{1}_{B_q \geq 2}$ on admissible paths, independently of the chosen admissible value of x_q. For $q < e$ the carry is zero.

Fix $x_{<i}$. The preceding identity determines γ_{i+e} from $y_0, \ldots, y_{i-1}$, hence from $x_{<i}$, independently of the intermediate admissible digits. Also $y_i \equiv u_0 x_i + R_i(x_{<i}) \pmod 3$, with $u_0 \in \{1, 2\}$. The output at position $i + e$ consequently has the form

$$x_{i+e} + u_0 x_i + Q_i(x_{<i}) \pmod 3. \tag{6}$$

At least one of $x_i = 0, 1$ makes the offset in (6) nonzero modulo 3, leaving at most one allowed value of x_{i+e}, regardless of the intermediate admissible digits.

Mark that future position as forced. A mark means that each subsequent admissible prefix on the branch has at most one child at the marked position; it does not specify a digit common to different branches. Maintain the set S of future marks. At a marked position there is at most one child. At an unmarked position with one child, retaining S suffices for an upper bound. At an unmarked position i with two children, one child retains S, while the other has the additional mark $i + e$, if that position is less than n. Additional restrictions may be discarded.

Backward induction now compares the number of completions with independent sets of the remaining distance-e graph, with positions in S required absent. At an unmarked vertex i, its deletion recursion is the sum of the counts without and with the additional mark $i + e$; at a marked vertex there is a single continuation. Requiring additional positions absent can only decrease the count. This proves $|W_n(r)| \leq H_n(e)$, starting with no marks.

The graph is a union of at most e paths, one in each residue class modulo e. A path of m vertices has F_{m+2} independent sets, and $F_{m+2} \leq \varphi^{m+1}$ by the Fibonacci recurrence. Multiplying these inequalities gives $H_n(e) \leq \varphi^{n+e}$. □

Lemma 4.2 (Valuation summation)**.** *For $Y \geq 3$,*

$$\sum_{\substack{N < N' \\ N, N' \in S_Y}} \varphi^{v_3(N'-N)} \ll Y^{2s}.$$

Proof. Put $K = \lfloor \log_3 Y \rfloor + 1$. For fixed $N \in S_Y$ and $1 \le e < K$,

$$\#\{N' \in S_Y : N' \equiv N \pmod{3^e}\} \le 2^{K-e} \le 2Y^s 2^{-e}.$$

No distinct pair has valuation at least K. Bounding the number of pairs of valuation exactly e by the number with valuation at least e, and using $|S_Y| \le 2Y^s$, proves

$$\sum_{N<N'} \varphi^{v_3(N'-N)} \le 2Y^{2s} \sum_{e\ge 1} (\varphi/2)^e \ll Y^{2s}.$$

□

Proposition 4.3 (Overlap bound)**.** *For $X \ge 1$, $Y \ge 3$ and any $G \subset S_Y$,*

$$\sum_{\substack{N<N' \\ N,N'\in G}} |A_N(X) \cap A_{N'}(X)| \ll X^g Y^{2s+g}.$$

Proof. For a distinct pair, put $r = N'/N$ and $e = v_3(N' - N) = v_3(r - 1) \ge 1$. If $M \in A_N \cap A_{N'}$, then $x = NM \le XY$ and $x, rx \in \mathcal{D}$. The assignment $M \mapsto x$ is injective. Taking $n = \lceil \log_3(XY + 1) \rceil$ and applying Lemma 4.1 gives

$$|A_N \cap A_{N'}| \le \varphi^{n+e} \ll (XY)^g \varphi^e.$$

Sum this using Lemma 4.2. □

5 Proof of the power lower bound

Proof of Theorem 1.1. Take $Y = c_0 X^\alpha$, where $c_0 > 0$ is a sufficiently small fixed constant. We have $0 < \alpha < 1/6$. To check the latter without numerical rounding, note that $s < 7/11$ since $2^{11} < 3^7$, and $g > 3/7$ since $\varphi^7 = 13\varphi + 8 > 27$. The ratio $(s - g)/(1 + g)$ is increasing in s and decreasing in g, so $\alpha < 8/55 < 1/6$. Consequently $L = \lfloor \log_3 X \rfloor \ge 3\lceil 2 \log_3 Y \rceil$ for large X. Also $X^s Y^{s-1} \to \infty$, since $s > 1/2$ and $\alpha < 1/6$.

Proposition 3.3 provides a set $G \subset S_Y$ with first moment at least $X^s Y^{2s-1}/64$. By Proposition 4.3, the sum of pairwise overlaps is at most $C_0 X^g Y^{2s+g}$. The ratio of this upper bound to the first-moment lower bound is

$$64 C_0 X^{g-s} Y^{1+g} = 64 C_0 c_0^{1+g}.$$

Choose c_0 so that this is at most $1/2$. Bonferroni's inequality gives

$$M_{\mathrm{C}}(X) \ge \left| \bigcup_{N\in G} A_N(X) \right| \ge \frac{1}{128} X^s Y^{2s-1} = \frac{c_0^{2s-1}}{128} X^\beta.$$

Finally, if $N, MN \in \mathcal{D}$ with $N > 0$, choose B with $3^B > \max(N, MN)$. For every sequence $(\epsilon_j) \in \{0,1\}^{\mathbb{N}_0}$,

$$x = N \sum_{j\ge 0} \epsilon_j 3^{Bj}$$

and Mx consist of nonoverlapping copies of the finite digit blocks for N and MN. Thus both belong to Σ. This embeds a scaled block Cantor set, of dimension $\log 2/(B \log 3) > 0$, into $C(1, M)$. Hence $M_{\mathrm{H}}(X) \ge M_{\mathrm{C}}(X)$, proving the second bound. □

6 A delayed digit equation

Fix $k \geq 0$ and put

$$m = 4k + 3, \qquad M = 23 \cdot 3^m - 20.$$

Suppose, for a contradiction, that $Ma = b$ with $a, b \in \mathcal{D}^+$. Write their finite ternary expansions as $a = \sum_{j\geq 0} a_j 3^j$ and $b = \sum_{j\geq 0} b_j 3^j$, extending both digit sequences by zeros. For negative indices set $a_j = 0$. The identity $b + 20a = 23 \cdot 3^m a$ gives the delayed digits $x_j = a_{j-m}$ and the recurrence

$$d_0 = 0, \qquad d_{j+1} = \frac{d_j + 20a_j + b_j - 23x_j}{3}. \tag{7}$$

Lemma 6.1. *Every d_j in (7) is an integer in $\mathcal{I} = \{-11, -10, \ldots, 10\}$, and $d_j = 0$ for all sufficiently large j.*

Proof. Set $f_h = 20a_h + b_h - 23a_{h-m}$. The finite identity $\sum_{h\geq 0} f_h 3^h = 0$ implies

$$d_j = 3^{-j} \sum_{h<j} f_h 3^h \in \mathbb{Z},$$

since the omitted terms are divisible by 3^j. This formula gives (7) and is zero once the partial sum includes every nonzero term. If $d_j \in \mathcal{I}$, its next numerator lies between -34 and 31. Since the numerator is divisible by 3, it lies between -33 and 30, so $d_{j+1} \in \mathcal{I}$. Induction starts at 0. □

Encode the last m input digits by the *binary* register

$$A_j = \sum_{i=0}^{m-1} a_{j-m+i} 2^i.$$

The oldest digit is in bit 0, so

$$x_j = A_j \bmod 2, \qquad A_{j+1} = \lfloor A_j/2 \rfloor + a_j 2^{m-1}. \tag{8}$$

This binary encoding is only a convenient label for a queue of digits. It is not the ordinary ternary value of that queue. The state space is

$$\{0, \ldots, 2^m - 1\} \times \mathcal{I}.$$

The initial state is $(0, 0)$; a finite witness gives a finite path back to $(0, 0)$ containing an input digit 1.

Lemma 6.2. *Any positive witness gives a path from $(0, 0)$ to $(0, -1)$. Consequently, no positive witness exists if there is a set of states containing $(0, -1)$, closed under all predecessors, and omitting $(0, 0)$.*

Proof. Because $M \equiv 1 \pmod 3$, the same power of 3 divides a and b. Dividing it out preserves their allowed digits, so we may assume $a_0 = b_0 = 1$. The first state after $(0, 0)$ is then $(2^{m-1}, 7)$, which is different from $(0, 0)$.

Consider the first subsequent return to $(0, 0)$. A transition into this state has $a_j = 0$ and $A_j = x_j \in \{0, 1\}$ by (8). Equation (7) gives $d_j = 23x_j - b_j$. If $x_j = 1$, this value is 22 or 23, outside $\mathcal{I}$. If $x_j = 0$, the predecessor is $(0, 0)$ or $(0, -1)$. The first possibility is excluded at the first return. Thus the path passes through $(0, -1)$. A predecessor-closed set containing that state must contain every earlier state on the path, including $(0, 0)$, giving the asserted contradiction. □

For use in an explicit certificate, all predecessors can be written without a search. For a current state (A, d) put $\alpha = \lfloor A/2^{m-1} \rfloor \in \{0, 1\}$. For each $x, \beta \in \{0, 1\}$ the potential predecessor is

$$A^- = 2A - \alpha 2^m + x, \qquad d^- = 3d - 20\alpha - \beta + 23x. \tag{9}$$

It is a valid predecessor exactly when $d^- \in \mathcal{I}$. Here α is the input digit of the forward transition and β is its output digit. Formula (9) enumerates all choices; automatically $0 \le A^- < 2^m$.

7 The predecessor certificate for all $k \ge 1$

Assume now $k \ge 1$. Define

$$T_j = \frac{16^j - 1}{5} \quad (j \ge 1), \qquad V = \frac{3T_{k+1} - 1}{2}. \tag{10}$$

The numbers T_j are integers satisfying $T_{j+1} = 16T_j + 3$; their binary expansions are repetitions of the block 0011 after padding to length $4j$.

The following finite sets specify the allowed carries in the certificate:

$$\begin{aligned}
Z &= \{-10, -9, -4, -3, -1\}, & O &= \{-8, -7, -5, -4, 10\},\\
P &= \{-2, -1, 1, 2, 7, 8, 10\}, & P^+ &= P \cup \{4, 5\},\\
Q &= \{-7, -6, -4, -3, 2, 3, 5, 6\}, & R &= \{-10, -9, 5, 6, 8, 9\},\\
S &= \{1, 2, 4, 5, 10\}, & U &= \{-8, -7, -5, -4\},\\
U^+ &= U \cup \{-6, -3\}, & W &= \{2, 3, 5, 6\},\\
W^+ &= W \cup \{4, 7\}, & H &= \{5, 6, 8, 9\},\\
I_0 &= \{-10, -9\}, & J_0 &= \{-11, -6, -5, -3, -2\},\\
K_0 &= \{-10, -9, -7, -6, 8, 9\}, & N &= U.
\end{aligned}$$

The carry-set symbols in this display are local to the certificate. Distinct row names identify different registers, even when their carry sets agree.

Table 1: Definition of $\mathcal{S}_m$. Empty index ranges are omitted.

Row	Index range	Register A	Carry set
Z_0		0	Z
O_0		1	O
P_j	$1 \le j \le k$	T_j	P
P_{k+1}		T_{k+1}	P^+
Q_j	$1 \le j \le k+1$	$2T_j$	Q
R_j	$1 \le j \le k$	$4T_j$	R
S_j	$1 \le j \le k$	$4T_j + 1$	S
U_j	$1 \le j < k$	$8T_j + 1$	U
U_k		$8T_k + 1$	U^+
W_j	$1 \le j < k$	$8T_j + 2$	W
W_k		$8T_k + 2$	W^+
H_j	$1 \le j \le k$	$T_{j+1} + 1$	H
I_0^{row}		$T_{k+1} - 1$	I_0
I_1^{row}		$V - 1$	I_0
J^{row}		V	J_0
K^{row}		$V + 1$	K_0
N^{row}		$2T_{k+1} - 1$	N

Proposition 7.1. *The set $\mathcal{S}_m$ defined in Table 1 is closed under all predecessors in* (9)*. It contains* $(0,-1)$ *and omits* $(0,0)$*. Its cardinality is* $38k+50$*.*

Proof. We first determine the most significant bit of every register. Put $X = 16^k \geq 16$, so $2^{m-1} = 4X$ and $2^m = 8X$. For $j \leq k$, $T_j < X/5$. All rows with most significant bit zero are bounded by

$$T_{k+1} + 1 = \frac{16X+4}{5} < 4X; \qquad 8T_j + 2 \leq \frac{8X+2}{5} < 4X.$$

The rows $Q_{k+1}, I_1^{\mathrm{row}}, J^{\mathrm{row}}, K^{\mathrm{row}}, N^{\mathrm{row}}$ have respective registers

$$\frac{32X-2}{5}, \quad \frac{24X-9}{5}, \quad \frac{24X-4}{5}, \quad \frac{24X+1}{5}, \quad \frac{32X-7}{5}.$$

Each lies in $[4X, 8X)$, so its most significant bit is one. This also verifies that every row defines a state in the prescribed space.

Table 2 gives the complete predecessor calculation. An unannotated destination means its whole carry set. A qualifier such as "base U" means the displayed smaller carry set, even when the destination row uses U^+. A dash denotes an empty image. Every entry allows both values $\beta = 0, 1$ and retains only $d^- \in \mathcal{I}$.

Table 2: All predecessor images from (9).

Current row	α	$x=0$	$x=1$
Z_0	0	$Z_0 : \{-10,-9,-4,-3\}$	O_0
O_0	0	—	P_1 (base P)
P_j, all j	0	Q_j	—
Q_j, $j \leq k$	0	R_j	S_j
Q_{k+1}	1	J^{row}	K^{row}
R_j	0	—	U_j (base U)
S_j	0	W_j (base W)	—
U_j, $j < k$	0	—	P_{j+1} (base P)
U_k	0	I_0^{row}	P_{k+1} (P^+)
W_j, all j	0	H_j	—
H_j	0	—	—
I_0^{row}	0	—	N^{row}
I_1^{row}	1	—	—
J^{row}	1	—	$U_k : \{-7,-6,-4,-3\}$
K^{row}	1	$W_k : \{3,4,6,7\}$	—
N^{row}	1	—	I_1^{row}

For the carry column, each entry is the finite calculation

$$\{3d - 20\alpha - \beta + 23x : d \text{ in the current carry set}, \ \beta \in \{0,1\}\} \cap \mathcal{I}.$$

The register column follows from $T_{j+1} = 16T_j + 3$ and the two boundary identities

$$4T_{k+1} - 2^m = V, \qquad 2V - 2^m = 8T_k. \tag{11}$$

For example, the row Q_{k+1} has register $2T_{k+1}$ and $\alpha = 1$. Its predecessor registers are $V, V+1$, and its carry images are J_0, K_0, respectively. The row J^{row} has $\alpha = 1$; for $x = 1$ it maps to register $8T_k + 1$ and carries $\{-7,-6,-4,-3\} \subseteq U^+$. These boundary images explain the extra carries in P^+, U^+, W^+.

Thus every possible predecessor lies in a listed row. This verifies closure for all $k \geq 1$ using fixed finite carry sets and the exact identities (10) and (11). No upper bound on k

is used. Since Z contains -1 and excludes 0, and no other row has register zero, the two required inclusions follow.

For the cardinality, the registers are distinct. In the interior,

$$T_j < 2T_j < 4T_j < 4T_j + 1 < 8T_j + 1 < 8T_j + 2 < T_{j+1},$$

and $T_{j+1}+1$ lies strictly between T_{j+1} and $2T_{j+1}$. At the final boundary, writing $T = T_{k+1} \geq 51$, the additional order is

$$T - 1 < T < T + 1 < V - 1 < V < V + 1 < 2T - 1 < 2T.$$

The register 1 lies below $T_1 = 3$. Summing the sizes of the carry sets in Table 1 now gives $38k + 50$. □

Lemma 6.2 and Proposition 7.1 exclude a positive witness for every $k \geq 1$.

8 The base case and the sporadic exceptional set

For $k = 0$ we have $m = 3$ and $M = 601$. The small certificate in Table 3 replaces Table 1.

Table 3: A predecessor-closed certificate of 65 states for $m = 3$.

Register A	Carry values
0	$\{-10, -9, -4, -3, -1, 3, 4, 6, 7, 8, 9\}$
1	$\{-8, -7, -6, -5, -4, -3, 10\}$
2	$\{-10, -9, 3, 4, 6, 7\}$
3	$\{-10, -9, -2, -1, 1, 2, 4, 5, 7, 8, 10\}$
4	$\{-11, -6, -5, -3, -2, 8, 9\}$
5	$\{-10, -9, -8, -7, -6, -5, -4, 8, 9\}$
6	$\{-7, -6, -4, -3, 2, 3, 5, 6\}$
7	$\{-10, -9, -8, -7, -5, -4\}$

Indeed, for each entry (A, d) and $x, \beta \in \{0, 1\}$, put

$$\alpha = \lfloor A/4 \rfloor, \qquad A^- = 2A - 8\alpha + x, \qquad d^- = 3d - 20\alpha - \beta + 23x.$$

Whenever $-11 \leq d^- \leq 10$, the value d^- occurs in the row A^- of Table 3. This is a direct check of 260 integer expressions. The table contains $(0, -1)$ and omits $(0, 0)$, so Lemma 6.2 also excludes a witness for $k = 0$.

Proof of Theorem 1.2. The certificates prove $M \notin \mathcal{V}$ for every $k \geq 0$. Since $M \equiv 1 \pmod 3$, it does not belong to $\mathcal{O}_3$: an element of $\mathcal{O}_3$ coprime to 3 would have to be congruent to 2 modulo 3. For $m \geq 3$,

$$2 \cdot 3^{m+2} < 23 \cdot 3^m - 20 < \tfrac{3}{2}\, 3^{m+3}.$$

These are the upper and lower endpoints of two consecutive intervals in the definition of $\mathcal{O}_\infty$, so $M \notin \mathcal{O}_\infty$. Hence $M \in \mathcal{F}$. The numbers $621 \cdot 3^{4k} - 20$ are distinct and unbounded, proving that $\mathcal{F}$ is infinite. □

9 An improved exponent without an epsilon loss

This section is new in version 2. The original power bound and the exceptional-family proof above are retained with their original numbering. Write $s = \log_3 2$, $\varphi = (1 + \sqrt{5})/2$, and $g = \log_3 \varphi$.

Theorem 9.1. *There are absolute constants $c_* > 0$ and X_0 such that*

$$M_{\mathrm{C}}(X) \geq c_* X^{\beta_*} \quad (X \geq X_0), \qquad \beta_* = \frac{(1+s)^2}{7-5s} = 0.691726633148796\ldots.$$

The same lower bound holds for $M_{\mathrm{H}}(X)$. No subtraction of an arbitrary positive epsilon from β_* is required.

9.1 A boundary estimate for the mean Fourier discrepancy

Use $F_l, f_l, \mathcal{D}_l$ from Section 2. Their second and fourth moments are 2^{-l} and $(3/8)^l$. Let T_D be all nonzero reduced fractions on $\mathbb{T}$ with denominator at most D coprime to 3, and put $S_D(L) = \sum_{z \in T_D} f_L(z)$ and $h = \lceil 2\log_3 D \rceil$. Multiplication by 3 permutes T_D, whose separation is at least D^{-2}. The sampling lemma gives, uniformly for $0 \leq m \leq h$,

$$\sum_{T_D} f_m^2 \ll D^2 2^{-m}, \qquad \sum_{T_D} f_m^4 \ll D^2 (3/8)^m.$$

Indeed $3^h < 3D^2$, so the degrees in both applications are $O(D^2)$. Set

$$a_* = \frac{3s-1}{4}, \quad A_* = \frac{2(3-s)}{3s-1}, \quad \eta_0 = 3-4s, \quad r_* = \frac{3-s}{2} = a_* A_*.$$

Here $4 < A_* < 6$, $0 < \eta_0 < s$, and $r_* > 1$. For $2h \leq L < 3h$ write $l = L - 2h$ and apply Holder's inequality with exponents $2, 4, 4$ to blocks of lengths h, h, l. This yields

$$S_D(L) \ll D^2 2^{-h/2} (3/8)^{(h+l)/4} \leq C D^{(3+s)/2} 3^{-a_* L}. \tag{12}$$

The last inequality uses $s - 1 < 0$ and $h \geq 2\log_3 D$. For $L \geq 3h$, dropping extra factors instead gives

$$S_D(L) \leq S_D(3h) \ll D^{\eta_0}. \tag{13}$$

For each fixed D, also $S_D(L) \to 0$: every nonzero fraction has a finite orbit under multiplication by 3, with all factors $|(1 + \exp(2\pi i z))/2| < 1$.

Let $S_Y = \{N \leq Y : N \in \mathcal{D},\ N \equiv 1 \pmod 3\}$ and $R_N(L) = \sum_{u=1}^{N-1} f_L(u/N)$.

Lemma 9.2. *There is an absolute C_* such that, for every fixed $H \geq 0$ and every integer choice $L = L(Y) \geq A_* \log_3 Y + H$,*

$$\limsup_{Y \to \infty} Y^{-s} \sum_{N \in S_Y} R_N(L) \leq C_* 3^{-a_* H}.$$

Proof. Fixing the high ternary digits shows that $T_Y(d) = \#\{N \in S_Y : d \mid N\} \leq 4(Y/d)^s$ for $1 \leq d \leq Y$: choose a low block with $3^j \leq d < 3^{j+1}$; its values lie in an interval of length less than d, hence contribute at most one multiple for each high block. Reducing u/N gives the exact identity

$$\sum_{N \in S_Y} R_N(L) = \sum_{\substack{2 \leq d \leq Y \\ (d,3)=1}} T_Y(d) \sum_{\substack{1 \leq u < d \\ (u,d)=1}} f_L(u/d).$$

In a fixed dyadic range $D/2 < d \leq D$, its contribution divided by Y^s is $O(D^{-s} S_D(L))$. One may take D to be powers of 2, with the final partial range bounded by $D < 2Y$. For ranges with $3h \leq L$, equation (13) supplies the summable majorant $C D^{\eta_0 - s}$. Each fixed range tends to zero, so dominated convergence makes their total $o(1)$. For all remaining ranges, $2h \leq L$ once Y is sufficiently large, because $A_* > 4$. Equation (12) gives

$$D^{-s} S_D(L) \ll D^{r_*} 3^{-a_* L} \leq 3^{-a_* H} (D/Y)^{r_*}.$$

The sum of $(D/Y)^{r_*}$ over these dyadic ranges is bounded absolutely. This proves the stated limsup. In particular the estimate asserts an arbitrarily small fixed proportion, not necessarily $o(Y^s)$ at fixed H. □

9.2 Affine tails and a fixed residue class

Lemma 9.3. *If $r \in 1+3\mathbb{Z}_3$, $r \neq 1$, $e = v_3(r-1)$, and $\xi \in \mathbb{Z}_3$, then*

$$\#\{z \in \mathcal{D}_m : [rz+\xi]_m \in \mathcal{D}_m\} \le H_m(e) \le \varphi^m (2/\varphi)^e,$$

where $[x]_m$ is the residue in $[0, 3^m)$ and $H_m(e)$ counts the independent sets of the graph on $0, \dots, m-1$ with edges of length e.

Proof. Write $r = 1 + 3^e u$, with u a unit. In the addition $z + 3^e uz + \xi$, let B_j be the sum of the incoming carry, the digit of ξ, and the shifted digit of uz. On an admissible path the outgoing carry is $\lfloor (B_j+1)/3 \rfloor$, independent of the admissible choice of z_j. Residues $B_j = 0, 1, 2 \pmod 3$ allow, respectively, two choices, only $z_j = 0$, or only $z_j = 1$. At position $i+e$, after fixing $z_{<i}$, the output digit is $z_{i+e} + u_0 z_i + Q_i(z_{<i}) \pmod 3$, with $u_0 \in \{1, 2\}$. Thus one of the two choices at i forces the digit at $i+e$, independently of intermediate admissible digits. Backward induction on the remaining positions bounds the tree by the independent-set recursion: at a free vertex one either removes that vertex, or removes it together with its future neighbor; an already forced vertex has only one branch. Discarding further restrictions only enlarges the bound. The graph splits into e chains, including empty chains. A chain of length l has $F_{l+2} \le 2\varphi^{l-1}$ independent sets, also for $l = 0$. Multiplying proves the second inequality. □

Fix $c = 112 = (11011)_3$ and $d = 128$ throughout the rest of the proof. Define

$$C_d(c) = \#\{a, b, a', b' \in \mathcal{D}_d : c(a-a') + (b-b') \equiv 0 \pmod{3^d}\}.$$

For every right-hand side the corresponding count is at most $C_d(c)$, by Cauchy–Schwarz applied to the autocorrelation of $h(z) = \#\{(a,b) \in \mathcal{D}_d^2 : ca + b \equiv z \pmod{3^d}\}$. Also $C_d(c) \ge (16/3)^d$.

The following finite certificate specifies the constant used here without a numerical spectral-radius approximation. On states $-56, \dots, 56$ put $T_{u,v} = w(3v-u)$, where

$$w(0) = 4, \quad w(\pm 1) = 2, \quad w(\pm 111) = 1, \quad w(\pm 112) = 2, \quad w(\pm 113) = 1$$

and all other weights vanish. These are precisely the digit-difference multiplicities. The state interval is closed under every integral transition. Starting with $v_0(u) = \mathbf{1}_{u=0}$, iterate $v_{j+1}(v) = \sum_u v_j(u) T_{u,v}$. Then $C = C_d(c) = \sum_u v_{128}(u)$, with unrestricted final carry, equals

$$C = 284803932576710365592543571298693298054902154739229935778\,5$$
$$0705669110397043217485591833069035\,92$$

where the two lines are concatenated decimal digits. The exact integer checks supplied with the article give

$$C^{1161} 3^{148608} \varphi^{491392} < 2^{937216}, \qquad 2^{20805} < 3^{13127}. \tag{14}$$

These comparisons require integers only: for $m = 491392$, write $\varphi^m = F_m \varphi + F_{m-1}$, multiply by $C^{1161}3^{148608}$ to obtain $A\varphi + B$, and check $D = 2^{937217} - A - 2B > 0$ and $D^2 > 5A^2$. The executable certificate recomputes the matrix both forwards and backwards and performs these comparisons exactly. Set $\lambda = C^{1/d}$ and $\tau = \log_3 \lambda$.

Lemma 9.4. *Let $Y = 3^k$, $k > d$, and $G \subset \{N \in \mathcal{D}_k : N \equiv 1 \pmod{3^d}\}$. For $Y < X$ put $A_N(X) = \{1 \le M \le X : M \equiv c \pmod{3^d},\ NM \in \mathcal{D}\}$. Then*

$$\sum_{N < N' \in G} |A_N(X) \cap A_{N'}(X)| \ll_d X^g Y^{\tau - g}.$$

Proof. For four prefixes N', x, N, y of length $m \geq d$ with low blocks $1, c, 1, c$ and $N'x \equiv Ny \pmod{3^m}$, adding four blocks of length d reduces the next congruence to

$$R + c(a - a') + (b - b') \equiv 0 \pmod{3^d}, \qquad R = (N'x - Ny)/3^m.$$

The quadratic terms vanish since $m \geq d$. There are at most C extensions. If the first distinct denominator digit occurs at e, there are at most 4^{e-d} common prefixes and two choices at e. Applying full blocks and bounding a remaining partial block trivially by 16^d proves that the number of length-k quadruples is $O_d(4^e\lambda^{k-e})$.

Choose $n \geq k$ with $X < 3^n \leq 3X$. For each pair, the map $M \mapsto [NM]_n$ is injective, since N is a unit. Conditional on a prefix x_0 of length k, its remaining digits satisfy an affine condition with ratio $r = N'/N$ and shift $\xi = (rx_0 - [rx_0]_k)/3^k$. Lemma 9.3 bounds these tails. Summing over e therefore gives

$$O_d\left(\varphi^{n-k}\lambda^k \sum_{e \geq d} \left(\frac{8}{\varphi\lambda}\right)^e\right).$$

The ratio is at most $3/(2\varphi) < 1$, since $\lambda \geq 16/3$. This proves the assertion, including arbitrarily close distinct pairs. □

9.3 The first moment and completion of the proof

Put

$$\alpha_* = \frac{1}{A_* - 1} = \frac{3s - 1}{7 - 5s}, \qquad Y = 3^k = 3^{\lfloor \log_3(c_0 X^{\alpha_*}) \rfloor},$$

where $c_0 > 0$ will be fixed before $X \to \infty$. The shell $\mathcal{S}_{k,d} = \{N \in \mathcal{D}_k : N \equiv 1 \pmod{3^d},\ N \geq 3^{k-1}\}$ has exactly $2^{-d-1}Y^s$ members, all in $[Y/3, Y/2)$. Choose a fixed H with $C_*3^{-a_*H} < 2^{-d-5}$, and then $c_0 \leq 3^{-(H+d+4)/(A_*-1)}$. For $L_* = \lfloor \log_3(XY/3) \rfloor - d$ one has $L_* - A_* \log_3 Y \geq H + 2$. Lemma 9.2 and Markov's inequality show that at least half the shell, denoted G, has $R_N(L_*) \leq 1/2$ for all sufficiently large X.

For $L_N = \lfloor \log_3(NX) \rfloor$, count multiples of N of the form $z = c + 3^d w$, $w \in \mathcal{D}_{L_N - d}$. Orthogonality, permutation of residues by 3^d, and monotonicity of f_l give at least

$$\frac{2^{L_N - d}}{N}(1 - R_N(L_N - d)) \geq 2^{-d-2}X^s N^{s-1}$$

such multiples. They are positive and less than NX, and z/N lies in the required residue class. Consequently

$$\sum_{N \in G} |A_N(X)| \gg_d X^s Y^{2s-1}. \tag{15}$$

The exact inequalities (14) imply $\alpha_* < \alpha_0 = 1161/5000$ and $g - s + \alpha_0(1 + \tau - g - 2s) < 0$. The coefficient $1 + \tau - g - 2s$ is positive, since $\tau \geq 4s - 1$ and $2s > g$. Hence

$$g - s + \alpha_*(1 + \tau - g - 2s) < 0.$$

Lemma 9.4 is thus $o(1)$ times (15). Bonferroni's inequality gives

$$M_{\mathrm{C}}(X) \geq \left|\bigcup_{N \in G} A_N(X)\right| \gg X^s Y^{2s-1} \gg X^{s+(2s-1)\alpha_*} = X^{\beta_*}.$$

The positive-witness inclusion proved above transfers the bound to M_{H}. This completes the proof of Theorem 9.1. The constants can be very small and the threshold very large; no practical search threshold or sharp asymptotic is asserted.

10 Verification and scope

For the new bound, `code/endpoint/run_checks.py` independently recomputes the exact 113-state energy forwards and backwards, checks the integer sign certificates, and tests block lifting, affine tails, and endpoint algebra. These are finite inputs and consistency checks; the limiting estimates are proved in Section 9. The underlying archived arguments had internal reviews; the new English section received a focused integration review. The retained version-1 proof bodies are not represented as freshly re-reviewed.

All estimates for the power lower bound, including the required uniformity and limiting arguments, are proved in the text. The finite-word estimate counts every locally admissible word; it does not assume an infinite continuation. The accompanying elementary program `verify_power_gain_lemmas.py` checks exact finite digital energies, the finite-word bound for unit ratios modulo 3^n through $n = 9$, and the stated divisor-cylinder and valuation-bucket bounds at finite scales. These checks supplement the proofs. They cannot establish their asymptotic conclusions by themselves.

The nonrepresentation proof consists of the delayed-digit argument and the explicit predecessor certificates. Its parameter-dependent part is a symbolic calculation, rather than an extrapolation from finitely many tested values of k. The program `verify_parametric_certificate.py` checks every carry image in Table 2. Splitting the combined rows into interior and boundary variants gives 18 symbolic rows and 384 choices of carry and binary digits. The register identities are checked as affine polynomials in formal variables $X = 16^k$ and $Y = T_j$, with exact rational coefficients. The most significant bit inequalities are proved in Proposition 7.1. The program also checks all 260 predecessor choices in the separate $k = 0$ certificate and makes redundant integer instantiations to detect indexing or transcription errors. Both programs use only the Python standard library and exact integer or rational arithmetic.

Theorems 9.1 and 1.2 give a power lower bound and an explicit infinite sporadic family, respectively. They do not give a full asymptotic, a density theorem, or a classification of all nonrepresentable multipliers. The known positive-witness construction is sufficient, not necessary, for positive Hausdorff dimension.

ChatGPT was used in developing and editing the arguments, implementing checks, and conducting internal AI-assisted reviews from several mathematical perspectives. Those reviews are not external peer review or proof-assistant verification. Authorship is attributed solely to Timofei Iuzhakov.

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